4×4 matrices, and ψ is the wave function which has four components. This equation is valid also when the speed of light is finite and it is thus relativistically correct.

The analytical solution of the Dirac wave equation for the energy a single electron is enclosed in a square root. Normally the negative sign is neglected for many such solutions. However, Dirac later recognized that also the solution with negative energy could have a physical meaning. The *hole theory* assumes that the negative energy levels of our universe are completely filled with all kind of particles at all possible quantum states. If sufficient energy is added (> twice the rest mass of the particle) to a point in space it is possible to "lift" such a particle from the negative part of our universe into our normal space. Behind it is left a hole, which to us looks like the same particle but with opposite charge − an anti-particle. A similar theory is used to explain the electrical behavior of semiconductors, see Ch 9.

When the hole theory is generalized, it follows that all particles have their corresponding anti-particles, hence all the particles mentioned have their *antiparticles* (designated by a bar above the particle symbol), except the photon and the mesons, who are their own antiparticles. We may think about *antimatter* as consisting of antiprotons and antineutrons in an antinucleus surrounded by antielectrons (i.e. positrons). Superficially, there would be no way to distinguish such antimatter from our matter (sometimes called *koino matter*). It has been proposed that the universe is made up of matter and antimatter as a requirement of the principle of *symmetry*. In that case some galaxies, which perhaps can be observed, should be made up of antimatter. When such antimatter galaxies (or material expelled from them) collide with koino matter galaxies, both types of matter are annihilated and tremendous amounts of energy released.

Some of the problems with the Dirac hole theory, like a possibly unlimited amount of particles with negative energy, is addressed by the Quantum Field Theory (QFT). However, that theory also seems to have its problems like an infinite amount of vacuum energy. QFT will not be used here and the interested reader is referred to modern text books in particle physics.

In order to reach the goal of a comprehensive, yet simple, theory of the composition of all matter, the properties of the neutrinos and the quark theory must be considered.

2.5. THE NEUTRINO

The neutrino is a fermion that plays an essential role in the models of elementary particles and in the theory of the formation and development of the universe. The existence of the electron neutrino was predicted by Pauli in 1927 but it was not proven until 1956 when Reines and Cowan detected them in experiments at the Savannah River (USA) nuclear reactor. Since neutrinos are emitted in the β^- decays following fission, nuclear reactors are the most intense neutrino sources on earth. The detector in the discovery experiments consisted of a scintillating solution containing cadmium surrounded by

photomultipliers to observe the scintillations which occurred as a consequence of the following reactions:

$$\overline{\nu} + {}^{1}\mathrm{H} \rightarrow \mathrm{n(fast)} + \mathrm{e}^{+} \tag{2.11}$$

$$\mathrm{e}^{+} + \mathrm{e}^{-} \rightarrow 2\gamma_1 \tag{2.12}$$

$$\mathrm{n(thermal)} + {}^{113}\mathrm{Cd} \rightarrow {}^{114}\mathrm{Cd} + \gamma_2 \tag{2.13}$$

The γ's emitted are of different energy; the γ_1 is 0.511 MeV, but γ_2 much higher. There is also a time lag between the γ's because of the time required for the fast neutrons to be slowed down to thermal energy. The detection system allowed a delay time to ascertain a relation between γ_1 and γ_2 (delayed coincidence arrangement). When the reactor was on, 0.2 cpm were observed, while it was practically zero a short time after the reactor had been turned off thereby demonstrating the formation of neutrinos during reactor operation.

Since the 1950s it has become clear that neutrinos exist as several types — called flavors. In β^- decay an "anti-neutrino" is formed, while a "neutrino" is emitted in β^+ decay. Both these neutrinos are now referred to as *electron neutrinos*, $\overline{\nu}_e$ and ν_e, respectively.

Other elemental particles are the pion and the muon. The pions formed in nuclear particle reactions are unstable and decay with a life-time of 3×10^{-8} s into a *muon* and a μ neutrino:

$$\pi^{\pm} \rightarrow \mu^{\pm} + \nu_\mu \tag{2.14}$$

The mass of the muon is 0.1135 u (105.7 MeV). The muon is also unstable and has a life-time of 2×10^{-6} s; it decays into an electron, an e neutrino and a μ anti-neutrino:

$$\mu^{\pm} \rightarrow \mathrm{e}^{\pm} + \nu_e + \overline{\nu}_\mu \tag{2.15}$$

A third type (flavor) of neutrino, the tau-neutrino, is also known. It also exists in normal and anti-particle forms. Tau-neutrinos are formed in the decay of tauons (a kind of very heavy leptons formed in high energy reactions) and designed as (ν_τ) and ($\overline{\nu}_\tau$).

In 1979 Reines, Sobel and Pasierb made new neutrino measurements with a detector containing heavy water so that the neutrinos would split the ^{2}H atom into two protons and a high energy electron, or convert it into two neutrons and a positron. The neutron will decay into a proton and an electron. Both reactions would only be sensitive to the ν_e; by measuring the neutron yield, the number of ν_e's could be calculated and compared to the calculated ν_e flux from the reactor. The two different decays could be followed by measuring the time delays between neutron capture and γ emission (see Ch. 9). The ratio of these two measurements was 0.43 ± 0.17.

In summary there are three flavors of neutrinos: (i) the electron neutrino, ν_e, which accompanies β decay, (ii) the muon neutrino, ν_μ, which accompanies pion decay, and (iii) the tau neutrino, ν_τ, which is only involved in very high energy nuclear reactions. The three flavors of neutrinos all have their anti particles. They have spin, no charge ("quark charge zero"), but possibly a small mass. They react very weakly with matter, the reaction cross section (cf. Ch. 11) being of the order of 10^{-43} cm^2, depending on neutrino energy.

Various attempts have been made to determine the neutrino rest mass, for example by measuring the decay of soft beta emitters like tritium, or double β decay as $^{82}Se \rightarrow {}^{82}Kr$. The present published value for the mass of the electron neutrino is 0.07 eV or 1.25×10^{-37} kg. The three flavors of neutrinos are in a steady exchange, ν_e to ν_μ to ν_τ etc, i.e. they oscillate between the various flavors. Because speed is converted back and forth to neutrino rest mass the three flavors propagate with different speed. The total flux of neutrinos is large at the earth's surface, on the average 6.5×10^{10} s^{-1} m^{-2}.

An international collaboration built a neutrino detection station, Gallex, in a rock facility in the Gran Sasso mountain in the Appenines, Italy, which operated between 1991 and 1997. Some ^{71}Ga atoms in 30.3 tons of gallium metal was reacting with solar neutrinos to form ^{71}Ge ($t_{½}$ 11.4 d) which was converted to the gaseous hydride, GeH_4, and counted in a proportional detector.

In 1987 a large underground neutrino detector near Fairport, Ohio, in a few seconds registered a sudden burst of 8 events. Taking into account that the normal background rate is about 2 events per day, which is believed to be caused by neutrinos produced in the sun's fusion reactions, this was an exceptional occurrence not only because of the event rate but also because the source was located outside our solar system and was a bright new supernova, SN1987A, appearing in the Large Magellanic Cloud. This was a lucky observation because the previous "near by" supernova was observed in 1604 by Johannes Kepler. The neutrino observation preceded the optical confirmation and it has been calculated that about 10^{58} neutrinos were released in the explosion.

A very large neutrino detector, OPERA, is operating at the Gran Sasso facility replacing the earlier Gallex. It has been used to measure the speed of neutrinos by time of flight from a pulsed neutrino generating target at CERN.

One of the most significant effects of the neutrino mass relates to the mass of the Universe. According to the Big Bang Theory of the origin of the Universe (see Ch. 12) there should be as many neutrinos as there are photons in the microwave background radiation remaining from the Big Bang, or about 100 million times as many neutrinos as other particles. If these neutrinos had a mass >10 eV they would constitute the dominant mass in the Universe. This would mean that there would be enough mass in our Universe for gravitational attraction eventually to overcome the present expansion, and consequently we would have a closed, or possibly pulsating universe, instead of a universe

which will continue to expand infinitely. Surprisingly, current observations of the most distant galaxies suggest an accelerating expansion of the Universe instead.

2.6. QUARKS AND THE STANDARD MODEL

All particles are considered to be possible states in which matter can condense. These states are related to the force that forms them. In this sense the solar system is a state of gravitational force, an atom is a state of electromagnetic force, and a nucleon is a state of the strong interaction force. A particle can represent a positive energy state of a system while its analog antiparticle represents the negative state of the same system. Some regular patterns have been found for the elementary particles which indicate that many of them in fact may only be exited states of the same particle, differing in quantum numbers such as spin (or "hyper charge"); in fact hundreds of such states are now known. For example, the neutron has a mass corresponding to 939 MeV and spin 1/2, and there is a baryon with mass 1688 MeV and spin 5/2 with all its other properties like those of the neutron: the heavier particle must be a highly excited state of the neutron.

Though many attempts have been made to unify all particles into one simple theory, this did not succeed until the quark theory was developed. To explain this we have to go back somewhat in time.

The spin of a charged particle leads to the formation of a *magnetic moment* directed along the axis of "rotation". It was discovered in the late 1930s that the magnetic moment of the proton spin ($M_p = 1.41 \times 10^{-26}$ J T^{-1}) is about 1/700 of the electron spin ($M_e = 9.27 \times 10^{-24}$ J T^{-1}), although theory predicts a ratio of 1/1836 ($= M_p/M_e$; see Ch. 6). Also, the neutron has a negative magnetic moment ($M_n = -0.97 \times 10^{-26}$ J T^{-1}). The only explanation scientists could offer for this deviation was that the proton is not an evenly charged rotating sphere, but contains some "internal electrical currents", and also the neutron must contain some internal charges, which balance each other to appear uncharged. Thus, it was doubtful that protons and neutrons were truly elementary.

Around 1960 Hofstadter and co-workers at the large Stanford Linear Accelerator Center (SLAC, Ch. 16) proved that both the proton and the neutron have an uneven internal nuclear charge density. This came from studies of the scattering of high energy electrons ($\sim$1 GeV) against protons and neutrons. It was suggested by Gell-Mann that this could mean that the proton and neutron were composed of smaller particles, with fractional charge and mass, which he called *quarks*. The intense search for such particles (leading to the discovery of many new "elementary" particles) culminated in the late 1970s in experiments in which still higher electron energies (4 – 21 GeV) were used and the energy and scattering angle of the electrons measured. These revealed that the nucleons had hard internal scattering centers with charge 1/3 that of the electron and mass 1/3 that of the nucleon. These particles, quarks, are held together by *gluons*, which are carriers of the nuclear force.